HCML Monitoring Portal: Server flow (one request through the BFF)

Browser
nginx portal-web
Fastify portal-bff
TTL cache in memory
Zabbix API JSON-RPC 2.0
1 GET /bff/api/sites Cookie: hcml_session=<jwt>
2 GET /api/sites Host with its port, X-Forwarded-For preserved
The plugin chain, in registration order
3 helmet: security headers CSP off: this process serves JSON only
4 CORS: one allowed origin (WEB_ORIGIN)
5 rate limit: per IP, before auth, so a flood is rejected before JWT work
6 compression hook: brotli / gzip
alt [rate limit exceeded]
7 429 rate_limited "the limit is N per minute"
onRequest: forgery check, then the auth guard
8 forgery check, unsafe methods only: a cookie POST needs X-HCML-CSRF: 1 and an Origin naming this host
9 verify the JWT: Bearer header, else the cookie. Never ?token=, not even on the stream
10 requiredRole(url): first match wins, anything unmatched needs viewer
alt [cookie POST without the header, or from another site]
11 403 csrf, before any token is checked
[no or bad token]
12 401 unauthorized: "Sign in to continue."
[role too low]
13 403 forbidden: names the role needed
14 cached("sites", 15 s, fn)
alt [fresh, within TTL]
15 the stored value
no Zabbix call at all
[another request is already fetching]
16 await the same promise
coalesced: twenty NOC screens cost one upstream read
[stale but inside staleMs]
17 the old value, now
18 one background refresh
a failed refresh keeps the old value and is logged, not thrown
[miss]
19 host.get · problem.get · trigger.get · item.get
20 raw JSON-RPC result
21 store, LRU, max 500 keys
The part Zabbix cannot do
22 resolveSite(): tag, then host name, then inventory, then group
23 roll hosts up per site, worst severity wins
24 200 application/json content-encoding: br
25 same, unbuffered
When Zabbix does not cooperate
26 token rejected → 503 zabbix_auth "fix the token", not an opaque 500
27 no answer in ZABBIX_TIMEOUT_MS → 503 zabbix_timeout
28 JSON-RPC error → 502 zabbix_error, carrying Zabbix's own words
the address in ZBX_URL goes to the log, never into the response body
The one write, for comparison
29 POST /api/problems/acknowledge, operator only the cookie plus X-HCML-CSRF: 1
30 event.acknowledge, on the SEPARATE write token
31 invalidate("problems", holdMs)
32 push the new list to every open screen at once
Browser
nginx portal-web
Fastify portal-bff
TTL cache in memory
Zabbix API JSON-RPC 2.0